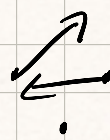
Last time:

Thrm: if $W \subset V$ and V is a vector space, then W is also a vector space if and only if its closed under scalar multiplication ($\forall a \in F$, $x \in W$ we have $ax \in W$) and it's closed under vector addition ($\forall x, y \in W$, $x+y \in W$)

Ex where this is false:

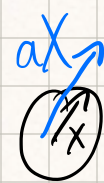
$V = \{\text{arrows in 2d}\}$ $F = \mathbb{R}$



$\boxed{W = \{\text{horizontal arrows}\} \text{ is a subspace}}$

$W = \{\text{arrows of length 1}\}$ is NOT a subspace

because for $a > 1$ $x \in W$ ax is not in W



Thrm: if F is any field then $F^n := \{a_1, a_2, \dots, a_n \mid a_i \in F\}$ is a vector space $\underbrace{\text{over } F}_{\text{"w/ field of scalars } F\text{"}}$

w/ scalar multiplication defined by

$$\underset{F}{x}(a_1, a_2, \dots a_n) = (xa_1, xa_2, \dots, xa_n)$$

and addition defined by

$$(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1+b_1, \dots, a_n+b_n)$$

PF: Check all the axioms in the def. of Vector space.

Ex: $F^1 = \{(a_1) \mid a_1 \in F\}$

$\hookrightarrow \mathbb{R}, \mathbb{C}$ are both vector spaces (over $\mathbb{R}$ and $\mathbb{C}$)

Ex: $\mathbb{C}^2 \ni (i,0), (1+i, 1-i), \ldots$

Ex: order matters $\mathbb{R}^2$ "points in the plane" $(1,2) \neq (2,1)$

Q: how much info do we need to know abt a linear func. to know its value on all inputs

Ex: $L: P_2 \to \mathbb{R}$ $L(x^2+2x) = 7$ is not enough

Ex: L is linear $L: \mathbb{R}^2 \to \mathbb{R}$ $x: (1,1)$ $y: (1,-1)$

suppose $L(x) = 2$, $L(y) = -3$

$L(a,b) \mid (a,b) = kx + ly$ for some $k, l \in \mathbb{R}$

$(a,b) = (2,3)$, does $(2,3) = kx + ly$?

$(2,3) = (k,k) + (l,-l) = (k+l, k-l)$

$$\begin{matrix} 2 = k+l \\ 3 = k-l \end{matrix} \quad k = 3+l \quad 2 = 3+2l \quad l = -\frac{1}{2}, \; k = \frac{5}{2}$$

So, $(2,3) = \frac{5}{2}x - \frac{1}{2}y$

$$L(2,3) = L\left(\frac{5}{2}x - \frac{1}{2}y\right) = \frac{5}{2}L(x) - \frac{1}{2}L(y) = 5 + \frac{3}{2} = \frac{13}{2}$$

This is enough